

DSA 210 — Introduction to Data Science

2025–2026 Spring Term — Project Proposal

Personalized Well-being Prediction from Sleep & Lifestyle Data

Motivation

Sleep is one of the most significant determinants of cognitive performance, mood, and physical health, yet its personal impact is poorly understood at the individual level. Commercial sleep trackers provide raw physiological metrics but offer no personalized interpretation or actionable insight. This project aims to build a dual-target model that predicts two meaningful personal outcomes — how rested one feels upon waking (morning feel rating) and how productive the following day turns out to be (daily productivity rating) — based on sleep metrics, lifestyle habits, and environmental conditions collected over 45 nights. A secondary goal is to identify the full set of personal conditions that simultaneously maximize both outcomes.

Data Sources & Collection

Apple Watch & Apple Ecosystem — exported via Apple Health XML and Screen Time:

Sleep stages (Light/Deep/REM/Awake), sleep duration, bedtime/wake time, Apple's proprietary nightly sleep score (objective sensor-derived quality index), resting heart rate, HRV, SpO₂, respiratory rate, daily steps, active calories, exercise minutes, and total/pre-sleep screen time across all Apple devices. Approximately 45 samples × ~15 sensor features.

Daily Self-Log — filled each morning and evening throughout the 45-day period:

Morning feel rating (1–10, first target), daily productivity rating (1–10, second target), caffeine count and time of last intake, alcohol consumption, nap duration, and last meal time before bed. 45 rows × 6 self-reported features.

External API Enrichment — retrieved automatically via OpenWeatherMap and a sunrise/sunset API:

Nighttime temperature, humidity, barometric pressure, and sunset time (used to derive natural light exposure and bedtime-to-sunset offset). Matched by date and location.

Analysis Plan

EDA and correlation analysis will be followed by hypothesis testing using non-parametric methods (Spearman correlation, Mann–Whitney U) suited to small samples. A dual-target predictive model (morning feel + daily productivity) will be built progressing from ridge regression and Lasso as primary approaches — appropriate for small samples — to tree-based comparisons (Random Forest, XGBoost) as secondary models, all evaluated via Leave-One-Out Cross-Validation (LOOCV) given $n = 45$. Including Apple's algorithmic sleep score as a feature allows the model to quantify the gap between objective sensor-based quality and subjective personal outcomes — itself a meaningful finding. SHAP feature importance outputs will then be used to derive a fully personalized daily routine recommendation covering optimal sleep duration, timing window, caffeine cutoff, meal timing, physical

activity, and alcohol habits that jointly maximize both targets.

Dataset Characteristics

Samples: 45 nights (April 1 – May 15, 2026)

Features: ~23 (sensor + Apple sleep score + self-log + environmental)

Targets: Morning feel rating (1–10) and daily productivity rating (1–10); both also used to derive a composite joint score for optimization

Evaluation: Leave-One-Out Cross-Validation (LOOCV) due to small sample size

Limitation: $n = 45$ is a small dataset; regularized linear models (Ridge, Lasso, ElasticNet) will be prioritized over tree-based methods to mitigate overfitting risk